

Network Incident Investigation and Remediation

Scenario

We are security analysts, supporting a large e-commerce client that reports suspected data leaks from a satellite office. Several customers have complained about phishing emails and vishing calls following conversations with Sales. We suspect the office has been compromised and that customer data is being exfiltrated and abused.

Our goal is to investigate, confirm the source and scope of the leak, and contain the impact. We will analyze the website, network traffic, and server logs, correlate findings across those data sources, and identify indicators of compromise. Throughout the engagement, we will coordinate with the client, provide regular updates, and deliver clear recommendations to mitigate the issue and prevent further damage.

Objectives

Investigate the incident.

4.8 Explain appropriate incident response activities.

4.9 Given a scenario, use data sources to support an investigation.

Analyze logs to gather more information about the incident. Look for any anomalies or suspicious activity that correlates with the information from the packet capture output.

2.4 Given a scenario, analyze indicators of malicious activity.

Identify the nature of the security incident. Determine the source of the breach, the affected systems or files, and the potential impact on the network.

4.8 Explain appropriate incident response activities.

Remediate the security incident, including removing any anomalous files from the system, modifying permissions to prevent unauthorized access, and implementing additional security measures as needed.

4.1 Given a scenario, apply common security techniques to computing resources.

4.5 Given a scenario, modify enterprise capabilities to enhance security.

Monitor the network for any further suspicious activity.

4.4 Explain security alerting and monitoring concepts and tools.

3.2 Given a scenario, apply security principles to secure enterprise infrastructure.

Exercise 1: Investigation

Task 1: Analyzing the pcap

1. On the [Security Onion 2.4.10](#) VM, on the desktop, open **traffic.pcap** with Wireshark.
2. In Wireshark, in the **Packet List** pane, review the pcap to analyze the network traffic.
3. List any external IP addresses you find:
75.30.5.55

Task 2: Reviewing the target machine

Displaying the IP address

1. Switch to the [Windows Server 2019](#) VM.
2. Display and document the IPv4 address of the Windows Server.
3. List the IP address of the Windows Server below:

Correct! 10.10.1.5

Displaying the active connections

1. Display the active connections and ports on which the computer is listening, then document any anomalous results.
2. List the port on which the attack is occurring below:

Correct! 1337

Checking for unauthorized accounts

1. Check for unauthorized accounts, then document any anomalous results.
2. List the anomalous user account below:

Correct! adm1nistrator

Checking for anomalous processes

1. Review the running applications, processes, and services, then document any anomalous results.
2. List the name of the malicious application below:

WinT00ls

Exercise 1: Knowledge Check

On which packet number does the attack start?

28

62

75

103

Correct!

What protocol is being used in the attack?

HTTP

SSHv2

ICMP

TCP

Correct!

Exercise 2: Log Analysis

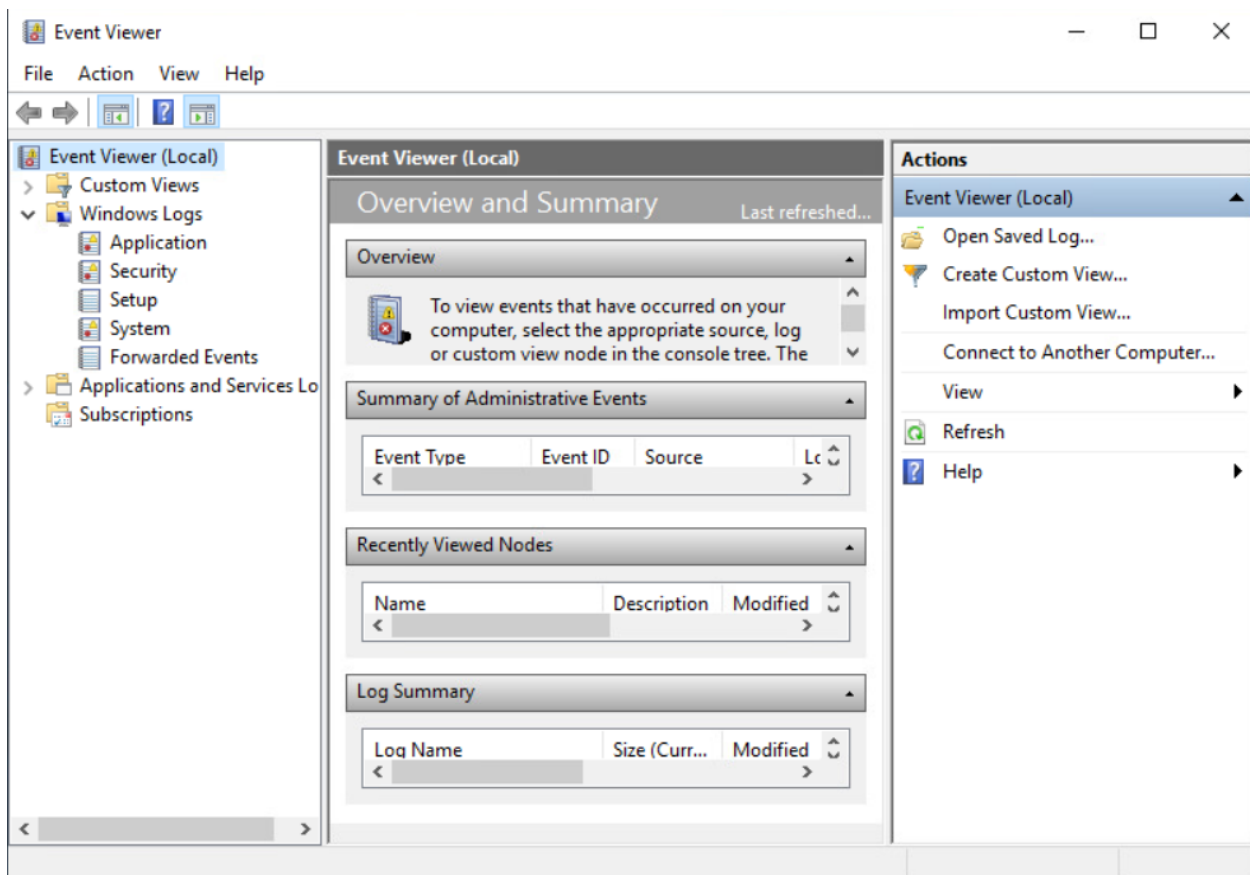
Task 1: Analyzing the Windows Event Viewer for user-creation events

1. Switch to the [Windows Server 2019](#) VM.
2. Examine the Windows security logs to determine if any unauthorized local accounts were created in the Windows Server 2019 machine.

3. In the Event Viewer, filter for event id **4720** to locate any logs regarding creating new user accounts.
4. If no user-creation events are found, be sure to check for any saved log to review.
5. Document your findings for use later in the lab.

Expand the following section for help completing this task

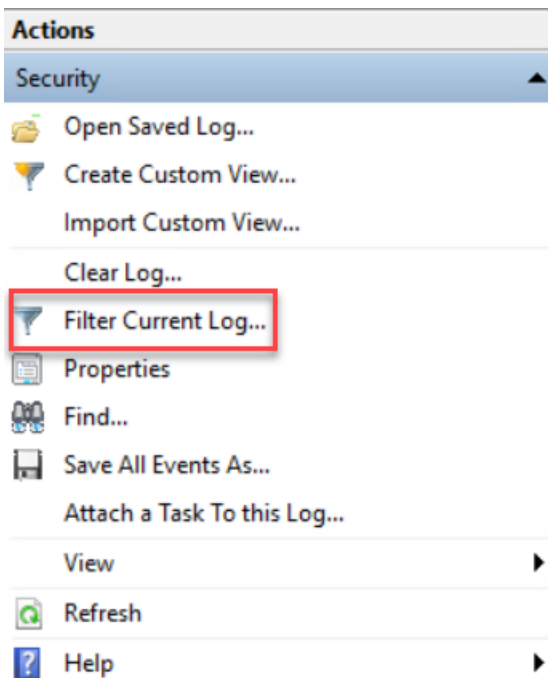
1. Right-click the **Start** button, then select **Event Viewer**, the Microsoft Windows Security Information and Event Management (SIEM) viewer.



2. Under **Event Viewer**, expand **Windows Logs**, then select **Security**.



3. In the **Actions** pane, select the **Filter Current Log**, to create a filter to show only the events related to User Account Management.



4. In the box, enter 4720, then select **OK**.

Filter Current Log ×

Filter XML

Logged: Any time

Event level: ☐ Critical ☐ Warning ☐ Verbose
☐ Error ☐ Information

☒ By log Event logs: Security

☐ By source Event sources:

Includes/Excludes Event IDs: Enter ID numbers and/or ID ranges separated by commas. To exclude criteria, type a minus sign first. For example 1,3,5-99,-76

<All Event IDs>

Task category:

Keywords:

User: <All Users>

Computer(s): <All Computers>

Clear

OK Cancel

5. If no result is returned, in the navigation pane, expand **Saved Logs**, and select **security_10-18**. Apply a filter for event ID 4720
6. Compare the account in the event to the anomalous user account found during the investigation to see if they match.
7. Document your findings for later use in the lab.

Resources

[Log Data: Key to Identifying Cybersecurity Threats](#)
[Event Viewer](#)

Task 2: Comparing your results to the pcap

In a previous task, the pcap was analyzed. Let's review the pcap details for specific details regarding the incident.

1. Switch to the [Security Onion 2.4.10](#) VM.
2. Select one of the packets that show the source IP address of 10.10.1.5 and the destination IP address of 75.30.5.55.
3. Examine the TCP Stream to determine the type of data that was sent from 10.10.1.5 to 75.30.5.55.

You should be able to clearly identify the data in the pcap.

4. Retrieve the IP address of the Security Onion VM and compare it to the pcap packet details.
5. List the Security Onion VM's primary IP address (ens32) below:

Correct! 10.10.1.55

Exercise 2: Knowledge Check

What type of encryption is being used on the compromised data?

SFTP

AES

None

RSA

Correct! The data is being sent in plaintext.

Which of the following, based on the exploit, is the malware type? Select the correct answer.

Spyware

Trojan

Rootkit

Worm

Correct! Based on the exploit the malware type is Spyware.

Exercise 3: Incident Identification

Task 1: Reviewing the target machine's findings

1. Review your notes to determine the data classification of the data sent from 10.10.1.5 to 75.30.5.55.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.10.1.55	192.168.10.22	TCP	74	54464 → 443 [SYN]
2	0.000186056	10.10.1.1	10.10.1.55	ICMP	102	Destination unreach
3	0.218699554	10.10.1.5	75.30.5.55	TCP	156	49673 → 1337 [PSH, Seq=1, Len=102]
4	0.218947133	75.30.5.55	10.10.1.5	TCP	60	1337 → 49673 [ACK, Seq=1, Len=0]
5	1.000514781	10.10.1.55	192.168.10.22	TCP	74	54472 → 443 [SYN]
6	1.000674945	10.10.1.1	10.10.1.55	ICMP	102	Destination unreach
7	1.234063328	10.10.1.5	75.30.5.55	TCP	161	49673 → 1337 [PSH, Seq=1, Len=102]
8	1.234261197	75.30.5.55	10.10.1.5	TCP	60	1337 → 49673 [ACK, Seq=1, Len=0]

Frame 3: 156 bytes on wire (1248 bits), 156 bytes captured (1248 bits) on interface ens32, id 0
Ethernet II, Src: Microsof_01:80:08 (00:15:5d:01:80:08), Dst: VMware_54:b5:21 (00:0c:29:54:b5:21)
Internet Protocol Version 4, Src: 10.10.1.5, Dst: 75.30.5.55
Transmission Control Protocol, Src Port: 49673, Dst Port: 1337, Seq: 1, Ack: 1, Len: 102
Data (102 bytes)

0000	00 0c 29 54 b5 21 00 15	5d 01 80 08 08 00 45 00	.)T ! E
0010	00 8e d5 f0 40 00 80 06	c9 15 0a 0a 01 05 4b 1e	 K
0020	05 37 c2 09 05 39 54 00	95 dc 42 5e 1b 9b 50 18	. 7 . . 9 T . . B ^ . . P
0030	20 14 11 24 00 00 44 41	56 45 20 4d 41 4e 44 49	. . S . . DA VE MANDI
0040	4e 4f 0a 73 65 78 3a 20	6d 0a 42 65 6e 6e 69 6e	NO-sex: m-Bennin
0050	67 74 6f 6e 20 42 6c 76	64 2e 2c 20 46 61 69 72	gton Blv d., Fair
0060	66 69 65 6c 64 2c 20 4e	43 0a 44 41 56 45 2e 4d	field, N C-DAVE.M
0070	41 4e 44 49 4e 4f 40 68	6f 74 6d 61 69 6c 2e 63	ANDINO@h otmail.c
0080	6f 6d 0a 33 30 34 2d 32	36 36 2d 30 32 36 31 0a	om-304-2 66-0261-
0090	36 34 32 2d 34 36 2d 36	31 39 32 0a	642-46-6 192-

2. Review your notes regarding the running applications and services to determine the data classification of the data sent from the Windows server.
3. Document the location of the malicious process below. Include only the path, not the file name.

Correct! C:\Users\admin\Downloads\dist

Exercise 3: Knowledge Check

Which of the following were compromised? (Check all that apply)

Customer addresses

Credit card numbers

Customer names

Social security numbers

Correct! Customer names, addresses, and social security numbers have been compromised in the attack.

Which of the following is the data classification that is being exfiltrated? Select the correct answer.

PCI

PII

PHI

PII, PHI, PCI

Correct! PII is the data classification that is being exfiltrated.

Exercise 4: Mitigation

Task 1: Mitigating the unauthorized user account

1. Disable the unauthorized user account you found on the Windows Server in a previous exercise.

Task 2: Mitigating the malicious application

1. Disable / end the malicious application running on the Windows Server.

You have successfully killed the malicious process.

Exercise 4: Knowledge Check

Which of the following is the Windows application that can be used to determine if the exploit resurfaces? Select the correct answer.

Explorer

Task Scheduler

Microsoft Antivirus Toolkit

Task Manager

Correct! Task Manager is the Windows application to use to determine if the exploit resurfaces.

Exercise 5: Monitoring

Task 1: Verifying the malicious application was terminated.

1. Verify that the malicious application was terminated and did not restart.

Expand the following section for help completing this task

1. In Windows PowerShell, run the following command to Verify that the malicious application was terminated.

`Get-Process -Name WinT00ls`

```
PS C:\Windows\system32>
PS C:\Windows\system32> get-process -name WinT00ls
get-process : Cannot find a process with the name "WinT00ls". Verify the process name and call the cmdlet again.
At line:1 char:1
+ get-process -name WinT00ls
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (WinT00ls:String) [Get-Process], ProcessCommandException
+ FullyQualifiedErrorId : NoProcessFoundForGivenName,Microsoft.PowerShell.Commands.GetProcessCommand
```

The resulting message indicates that the process was not found to be running.

Task 2: Verifying the malicious packets transmissions has been terminated.

1. Verify the malicious packets transmissions has been terminated by recapturing network traffic and reviewing the pcap.

Expand the following section for help completing this task

1. Switch to the [Security Onion 2.4.10](#) VM.
2. In Wireshark, on the menu, select **Capture > Start**.

Allow the capture to run for approximately 10 seconds, then select **Stop**.

3. Review the pcap to verify that there are no packets that have **75.30.5.55** as the destination.

No.	Time	Source	Destination	Protocol	Length	Info
	002719704	10.10.1.55	192.168.10.22	TCP	74	32774 → 9200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
Number	002796436	10.10.1.1	10.10.1.55	ICMP	102	Destination unreachable (Host unreachable)
	9 0.121951182	10.10.1.55	192.168.10.22	TCP	74	32790 → 9200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
	10 0.122167812	10.10.1.1	10.10.1.55	ICMP	102	Destination unreachable (Host unreachable)
	11 0.123992364	10.10.1.55	192.168.10.22	TCP	74	32802 → 9200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
	12 0.123187834	10.10.1.1	10.10.1.55	ICMP	102	Destination unreachable (Host unreachable)
	13 0.123910057	10.10.1.55	192.168.10.22	TCP	74	32812 → 9200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
	14 0.123991988	10.10.1.1	10.10.1.55	ICMP	102	Destination unreachable (Host unreachable)
	15 0.124653620	10.10.1.55	192.168.10.22	TCP	74	32814 → 9200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
	16 0.124742310	10.10.1.1	10.10.1.55	ICMP	102	Destination unreachable (Host unreachable)
	17 1.003986601	10.10.1.55	192.168.10.22	TCP	74	32822 → 9200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
	18 1.004312496	10.10.1.1	10.10.1.55	ICMP	102	Destination unreachable (Host unreachable)
	19 1.005241133	10.10.1.55	192.168.10.22	TCP	74	32830 → 9200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
	20 1.005342186	10.10.1.1	10.10.1.55	ICMP	102	Destination unreachable (Host unreachable)
	21 1.006098298	10.10.1.55	192.168.10.22	TCP	74	32836 → 9200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
	22 2.067411853	10.10.1.55	192.168.10.22	TCP	74	[TCP Retransmission] 32836 → 9200 [SYN] Seq=0 Win=64240
	23 2.621726894	10.10.1.55	192.168.10.22	TCP	74	32842 → 9200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
	24 2.621949590	10.10.1.1	10.10.1.55	ICMP	102	Destination unreachable (Host unreachable)
	25 2.623766833	10.10.1.55	192.168.10.22	TCP	74	32846 → 9200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
	26 2.623900826	10.10.1.1	10.10.1.55	ICMP	102	Destination unreachable (Host unreachable)
	27 2.624582465	10.10.1.55	192.168.10.22	TCP	74	32850 → 9200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
	28 2.624677711	10.10.1.1	10.10.1.55	ICMP	102	Destination unreachable (Host unreachable)
	29 2.625302203	10.10.1.55	192.168.10.22	TCP	74	32864 → 9200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
	30 2.625398613	10.10.1.1	10.10.1.55	ICMP	102	Destination unreachable (Host unreachable)
	31 4.116424944	10.10.1.55	192.168.10.22	TCP	74	[TCP Retransmission] 32836 → 9200 [SYN] Seq=0 Win=64240
	32 5.121313857	10.10.1.55	192.168.10.22	TCP	74	32876 → 9200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
	33 5.121523718	10.10.1.1	10.10.1.55	ICMP	102	Destination unreachable (Host unreachable)
	34 5.122411503	10.10.1.55	192.168.10.22	TCP	74	32884 → 9200 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
	35 6.163425114	10.10.1.55	192.168.10.22	TCP	74	[TCP Retransmission] 32884 → 9200 [SYN] Seq=0 Win=64240
	36 8.148421594	10.10.1.55	192.168.10.22	TCP	74	[TCP Retransmission] 32836 → 9200 [SYN] Seq=0 Win=64240
	37 8.211413772	10.10.1.55	192.168.10.22	TCP	74	[TCP Retransmission] 32884 → 9200 [SYN] Seq=0 Win=64240
	38 12.243409671	10.10.1.55	192.168.10.22	TCP	74	[TCP Retransmission] 32884 → 9200 [SYN] Seq=0 Win=64240
	39 13.459395575	10.10.1.55	192.168.10.22	TCP	74	51268 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK Pi
	40 16.532424923	10.10.1.55	192.168.10.22	TCP	74	[TCP Retransmission] 32836 → 9200 [SYN] Seq=0 Win=64240
	41 17.555384812	VMware_5b:f8:e8	VMware_54:b5:21	ARP	42	Who has 10.10.1.1? Tell 10.10.1.55
	42 17.555604675	VMware_54:b5:21	VMware_5b:f8:e8	ARP	60	10.10.1.1 is at 00:0c:29:54:b5:21
	43 20.627414858	10.10.1.55	192.168.10.22	TCP	74	[TCP Retransmission] 32884 → 9200 [SYN] Seq=0 Win=64240
	44 32.015418098	10.10.1.55	192.168.10.22	TCP	74	[TCP Retransmission] 32836 → 9200 [SYN] Seq=0 Win=64240
	45 37.012403655	10.10.1.55	192.168.10.22	TCP	74	[TCP Retransmission] 32884 → 9200 [SYN] Seq=0 Win=64240
	46 45.715400708	10.10.1.55	192.168.10.22	TCP	74	[TCP Retransmission] 51268 → 443 [SYN] Seq=0 Win=64240
	47 51.878995685	10.10.1.55	192.168.10.22	TCP	74	54512 → 3000 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK
	48 51.879234530	10.10.1.1	10.10.1.55	ICMP	102	Destination unreachable (Host unreachable)
	49 65.171508439	10.10.1.55	192.168.10.22	TCP	74	[TCP Retransmission] 32836 → 9200 [SYN] Seq=0 Win=64240